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Reliable Policy Transfer for Safety-Aware End-to-End Driving with Deep Reinforcement Learning

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https://github.com/szu-ai/safe-driving-drl/

Abstract

End-to-End (E2E) Reinforcement Learning (RL) for autonomous driving still struggles with safety and generalization under distribution shift, as perception-heavy encoders, sparse rewards, and ad hoc uncertainty handling yield brittle closed-loop behavior. This work introduces a unified Deep RL (DRL) framework built around a control-layer reliability interface where the same uncertainty signal informs relational attention, gates policy entropy, and regularizes transfer alignment. An ego-centric relational graph encodes agent influence via uncertainty-weighted attention over kinematics, lane geometry, and semantics, producing a compact control state. A multi-objective differentiable reward shapes safety, progress, and comfort with an uncertainty term. Aleatoric and epistemic uncertainty, captured through per-edge heteroscedastic variance and a critic ensemble, modulate policy entropy for risk-aware exploration. A causal-semantic transfer objective aligns actions, attention, and uncertainty statistics across domains with meta-learned initialization for few-shot adaptation. In closed-loop urban driving across varied towns, traffic, and weather, the framework improves success rate, reduces infractions, and achieves higher time-to-conflict combined with lower lateral deviation over strong baselines.

1. Introduction

Autonomous driving has advanced from modular pipelines to End-to-End (E2E) systems coupling scene understanding with control. Recent works demonstrate strong open-loop accuracy and scalable architectures [5, 10, 11, 14], yet robustness under distribution shift, dense traffic, and adverse weather remains limited [3]. Benchmarks emphasize online metrics like success rate and infractions per kilometer, revealing gaps not captured by open-loop evaluation [15]. Common E2E limitations include: state encoders compress perception without causal relations or confidence [9], rewards are sparse or discontinuous in safety-critical control [18], uncertainty is inconsistently estimated and rarely regulates exploration [1], and transfer emphasizes perception over control-level reasoning [4, 13, 20].

This paper addresses a central question: *how can safety-aware E2E driving policies be reliably transferred under distribution shift by aligning causality and uncertainty at the control layer?* A unified DRL framework is introduced with four innovations sharing a single control-layer reliability interface, where the same uncertainty signal informs relational attention, gates policy entropy, and regularizes transfer alignment. An ego-centric relational representation weights interactions by distance and uncertainty for influence modeling. A differentiable multi-objective reward provides feedback for safety, progress, and comfort. Joint aleatoric-epistemic estimation couples to the policy via entropy gating, aligning exploration with confidence [20]. A causal-semantic transfer objective aligns action distributions, attention, and uncertainty statistics across domains [13], with meta-learned initialization for adaptation [21].

The method instantiates an ego-node relational graph that aggregates edge features, lane geometry, and aleatoric variance via uncertainty-weighted attention, yielding a compact control state. Decision-time uncertainty is decomposed into aleatoric and epistemic components [9], with a critic ensemble stabilizing confidence [1]; a calibrated summary modulates policy entropy, reducing risk under low confidence and restoring exploration otherwise. For transfer, a regularizer combines policy Kullback-Leibler (KL) divergence, Maximum Mean Discrepancy (MMD) alignment, and moment matching of uncertainty statistics [13, 20], paired with a Model-Agnostic Meta-Learning (MAML) initialization for few-shot adaptation [21].

The framework is validated in closed-loop urban driving

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across diverse towns, traffic densities, and weather regimes [15]. Experiments show consistent gains in success rate and infractions per kilometer over strong E2E baselines [11], with higher time-to-conflict and improved lateral adherence [4]. Ablations confirm uncertainty-weighted attention and entropy gating are essential for robust safety, while the differentiable reward enhances optimization stability and transfer [26]. Causal-semantic alignment narrows the generalization gap across domains, enabling few-shot adaptation without sacrificing comfort or efficiency [13, 20].

This framework offers the following key contributions:

- Ego-centric relational state that fuses semantics and geometry into a compact control representation (Sec. 3.3).
- Differentiable reward that optimizes safety, progress, and comfort through uncertainty-aware shaping (Sec. 3.4).
- Unified uncertainty interface coupling relational attention, entropy-gated exploration, and transfer alignment via meta-initialized few-shot adaptation (Secs. 3.5–3.6).

2. Related Work

E2E driving with RL has advanced rapidly, but still struggles with state representation, reward shaping, uncertainty, and policy transfer, discussed in the following subsections.

2.1. Causal State Representation

Early E2E stacks emphasize perception and fusion but leave the control layer unaware of which entities causally influence the ego vehicle [5, 10]. Formulations such as UniAD and ThinkTwice improve task coupling, yet interactions are encoded through global feature tensors rather than explicit ego-centric relations [11, 14]. Vectorized and Bird's-Eye-View (BEV) representations enhance efficiency and geometry alignment but provide limited interpretability of edge-wise influences tied to safety outcomes [16, 22], and task decoupling does not expose uncertainty-weighted attention to prioritize interactions by distance and confidence [5, 13].

World models and language-augmented planners enrich semantics but emphasize capacity over compact, control-ready causal structure [23–25], and bias analyses reveal entanglement between geometry and control supervision [12]. Graph-style reasoning appears in prediction and planning, yet a unified state fusing lane geometry, goal progress, ego dynamics, and uncertainty into a compact vector remains uncommon [4, 6], motivating an ego-node graph encoding uncertainty-weighted structure for explicit policy learning.

2.2. Multi-Objective Reward Shaping

Reward shaping often relies on sparse or thresholded penalties, causing discontinuities and weak gradients [8, 18], and progress-only rewards encourage brittle behaviors in dense traffic [14, 15]. Richer supervision via prediction and planning heads was introduced, yet control objectives remain under-specified for smoothness and comfort [4, 11], while imitation-augmented methods inherit suboptimal biases and lack risk sensitivity under distribution shift [8, 12].

Differentiable multi-objective formulations improve stability when safety, progress, and smoothness are jointly optimized, though uncertainty is rarely integrated into reward terms [4, 27], and hierarchical approaches require domain-stable shaping to generalize across towns and weather [21, 23]. These gaps motivate differentiable shaping where the uncertainty signal is shared with attention and entropy gating rather than treated as a standalone term [15, 27].

2.3. Uncertainty Estimation for Exploration

Uncertainty-aware driving has explored distributional critics and ensembles to capture predictive dispersion, yet most systems treat aleatoric and epistemic components separately rather than jointly at decision time [1, 9]. Continual learning frameworks incorporate confidence but rarely propagate uncertainty through the scene representation used by the controller [5, 19]. Without uncertainty-weighted attention, policies risk overreacting to noisy detections or underweighting high-risk actors [10, 16], and RL entropy is scheduled globally rather than gated by confidence, weakening the link between risk and exploration [9, 17].

Safety gains have been reported when uncertainty modulates prediction and planning heads, though consistent connections to control entropy remain limited in E2E stacks [4, 24]. These gaps motivate a unified mechanism that jointly estimates both uncertainty types, routes them through the ego-centric graph, and gates policy entropy to align risk tolerance with confidence across the control-layer interface.

2.4. Semantic Alignment in Policy Transfer

Policy transfer often adapts perception features while leaving causal cues and uncertainty statistics misaligned across domains [13, 20]. Evaluations show overfitting to training towns or weather, with limited attention consistency in new scenes [11, 15]. Language-based adaptation and decoupling improve portability but do not explicitly align relational attention tied to safety-critical interactions [13, 20]. Meta-learning accelerates adaptation yet is rarely combined with uncertainty statistics to guide updates [21, 23], and causal consistency is better promoted by matching policy distributions and attention statistics across domains rather than pixel or BEV features [5, 16]. Uncertainty statistics are transferable signals that should be regularized to stabilize exploration and comfort after domain shift [9, 19].

Robotics studies indicate that semantics and uncertainty-informed transfer reduce drift and improve safety across simulators, motivating analogous designs in driving [2, 7]. These insights guide a causal-semantic transfer objective aligning actions, attention, and uncertainty statistics at the control layer for reliable few-shot adaptation.

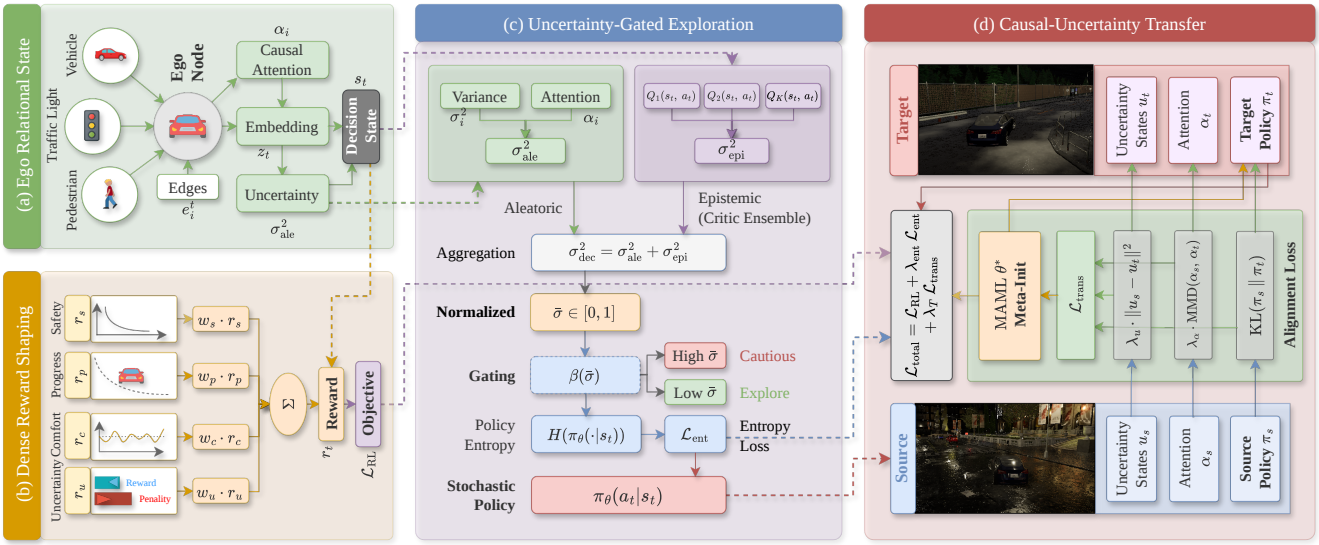


Figure 1. Unified framework: (a) relational state with uncertainty-weighted attention; (b) multi-objective reward shaping; (c) joint aleatoric–epistemic uncertainty gating of policy entropy; (d) causal-semantic transfer via KL, MMD, uncertainty matching, and MAML.

3. Unified End-to-End Policy Transfer

This section introduces a unified DRL framework for safety-aware autonomous driving built around a single control-layer reliability interface. Sec. 3.1 formalizes the MDP and Sec. 3.2 defines the closed-loop constraints; Sec. 3.3 presents an ego-centric relational graph producing a compact decision state; the normalized uncertainty signal $\bar{\sigma}$ calibrates the reward (Sec. 3.4), gates policy entropy (Sec. 3.5), and regularizes transfer alignment (Sec. 3.6), as shown in Fig. 1. The critic ensemble and MAML operate only during training; at inference, the controller uses a single actor pass with top-K relational aggregation.

3.1. Preliminaries

Closed-loop driving is modeled as an MDP $\mathcal{M} = (\mathcal{S}, \mathcal{A}, r, P, s_0, \gamma)$ with time-discretized control. The state space $\mathcal{S}$ contains ego-centric scene descriptions at time t, $s_t \in \mathcal{S}$. The action space $\mathcal{A}$ comprises continuous control commands $a_t = [a_t^{\text{thr}}, a_t^{\text{brk}}, a_t^{\text{str}}] \in \mathcal{A}$ for throttle, brake, and steering. The transition dynamics $P : \mathcal{S} \times \mathcal{A} \to \Delta(\mathcal{S})$ define $s_{t+1} \sim P(\cdot \mid s_t, a_t)$, capturing vehicle kinematics and interactions with other agents. The reward function $r : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \to \mathbb{R}$ evaluates safety, progress, smoothness, and uncertainty at each transition (s_t, a_t, s_{t+1}). The process starts at $s_0 \in \mathcal{S}$ and uses a discount factor $\gamma \in [0, 1)$.

A stochastic policy $\pi_\theta : \mathcal{S} \to \Delta(\mathcal{A})$ produces $a_t \sim \pi_\theta(\cdot \mid s_t)$ and is trained to maximize the discounted return

$$\max_\theta \; J(\theta) = \mathbb{E}_{\pi_\theta, P}\left[\sum_{t=0}^{\infty} \gamma^t \, r(s_t, a_t, s_{t+1})\right], \qquad (1)$$

where the expectation is over $a_t \sim \pi_\theta(\cdot|s_t)$ and $s_{t+1} \sim P(\cdot|s_t, a_t)$. Soft value functions V^π and Q^π support entropy-regularized actor-critic training, with temperature adjustment and target networks for stable updates.

For each entity i, relative kinematics are Δp_i (ego-frame position) and Δv_i (relative velocity), semantic class is c_i (vehicle, pedestrian, traffic light), and local lane features are κ_i (curvature, heading offset). Per-entity aleatoric variance σ_i^2 and decision-time uncertainty are decomposed into σ_{ale}^2 and σ_{epi}^2, aggregated into normalized summary $\bar{\sigma} \in [0, 1]$ that gates policy entropy. Lane adherence uses lateral deviation d_L and a context-dependent membership $\mu_A \in [0, 1]$ that tightens admissible deviation in fragile regimes.

3.2. Problem Formulation

The closed-loop task is to follow a route $\mathcal{R} = \{r_j\}_{j=1}^N$ along lane centerlines while interacting with agents and obeying traffic rules. At each step t, perception yields o_t, and a decision state $s_t \in \mathcal{S}$ is constructed for control as defined in Sec. 3.1. The policy samples $a_t = [a_t^{\text{thr}}, a_t^{\text{brk}}, a_t^{\text{str}}]$, the environment transitions to $s_{t+1} \sim P(\cdot \mid s_t, a_t)$, and the reward $r(s_t, a_t, s_{t+1})$ provides dense feedback. Control ensures adherence, avoids collisions, respects signals, and maintains comfort under partial observability.

Lane adherence is enforced on the active segment (r_j, r_{j+1}) using the lateral point-to-line distance

$$d_L(p_t; r_j, r_{j+1}) = \frac{\|(r_{j+1} - r_j) \times (p_t - r_j)\|_2}{\|r_{j+1} - r_j\|_2}, \qquad (2)$$

where p_t denotes the ego position at time t in the map frame. Admissible deviation is tightened in fragile contexts through a membership-controlled corridor,

$$d_L \le \epsilon(\mu_A), \quad \epsilon(\mu_A) = \epsilon_{\min} + (\epsilon_{\max} - \epsilon_{\min})\,\mu_A, \; (3)$$

with $\mu_A \in [0,1]$ computed from contextual factors such as traffic density, reduced visibility, or sharp curvature.

Progress is measured along the route tangent. Let s_t^{arc} denote the arc-length coordinate of p_t on $\mathcal{R}$, and let $\Delta s_t = s_{t+1}^{\text{arc}} - s_t^{\text{arc}}$ represent the step progress. Safety requires maintaining sufficient separation from the nearest entity $i^\star$,

$$\text{dist}(p_t,\, p_t^{(i^\star)}) \;\geq\; d_{\min}, \tag{4}$$

and a minimum time-to-collision (TTC) threshold, as $\tau_t = \frac{\|\Delta p_{i^\star}\|_2}{\max\left(\epsilon_v,\, (-\Delta p_{i^\star} \cdot \Delta v_{i^\star})_+\right)} \geq \tau_{\min}$, where $(x)_+ = \max(x, 0)$ and $\epsilon_v > 0$ prevents degeneracy at low speeds. Signal obedience and stopping-line compliance are incorporated as smooth reward penalties to avoid discontinuities.

Comfort is enforced by limiting jerk and steering rate. Let $j_t = \|a_t^{\text{veh}} - a_{t-1}^{\text{veh}}\|_2 / \Delta t$ be longitudinal jerk from ego acceleration a_t^{veh}, and $\dot{\delta}_t = (a_t^{\text{str}} - a_{t-1}^{\text{str}})/\Delta t$ the steering rate. Comfort bounds require $j_t \leq j_{\max}$ and $|\dot{\delta}_t| \leq \dot{\delta}_{\max}$, implemented as differentiable shaping terms; action bounds and actuator limits are enforced by output squashing.

The objective is to learn a policy that maximizes closed-loop performance. A Lagrangian relaxation combines dense reward r_{d} with smooth penalties $\phi_{\{\cdot\}}$. Let $\boldsymbol{\lambda} = [\lambda_L, \lambda_S, \lambda_C]^\top \in \mathbb{R}^3_{\geq 0}$ denote nonnegative multipliers for lane adherence, safety, and comfort, and $\boldsymbol{\phi}_t = \left[\phi_L(d_L, \mu_A),\; \phi_S(\text{dist}, \tau_t),\; \phi_C(j_t, \dot{\delta}_t)\right]^\top$ stack penalties at time t, where ϕ_L penalizes lateral deviation d_L with membership μ_A, ϕ_S penalizes safety violations via distance dist and time-to-conflict τ_t, and ϕ_C penalizes comfort violations through jerk j_t and steering rate $\dot{\delta}_t$. The objective is

$$\max_\theta \; \mathbb{E}\left[\sum_{t\geq 0} \gamma^t \Big(r_{\text{d}}(s_t, a_t, s_{t+1}) - \boldsymbol{\lambda}^\top \boldsymbol{\phi}_t\Big)\right]. \tag{5}$$

Sec. 3.4 specifies r_{d} and the differentiable penalties to avoid sparse events and stabilize gradients across domains.

This formulation raises four interconnected design questions, each addressed by one module: (1) *causal ego-centric state* to identify safety-relevant entities (Sec. 3.3); (2) *differentiable multi-objective reward* sharing $\bar{\sigma}$ with attention and entropy gating (Sec. 3.4); (3) *joint uncertainty estimation* propagated through the graph to gate exploration (Sec. 3.5); and (4) *causal-uncertainty transfer* aligning policy, attention, and confidence across domains (Sec. 3.6).

3.3. Causal State Representation

This subsection instantiates $s_t \in \mathcal{S}$ by exposing entities causally influencing control at time t, as formalized in Secs. 3.1–3.2. Prior fusion encoders compress observations into global tensors, leaving the policy unaware of priorities or confidence, hampering attribution and safety-driven gradients. The goal is an *ego-centric, compact, differentiable state design* integrating relational cues, ego dynamics, lane geometry, progress, and uncertainty.

Let $\mathcal{E}_t = \{e_i^t\}_{i=1}^M$ denote directed edges from scene entities into the ego node. Each edge carries a compact feature,

$$\mathbf{e}_i^t = \left[\Delta p_i,\; \Delta v_i,\; c_i,\; \kappa_i,\; \sigma_i^2\right], \tag{6}$$

where Δp_i and Δv_i are relative position and velocity in the ego frame, c_i is a semantic code (vehicle, pedestrian, signal), κ_i summarizes local lane geometry (curvature, heading offset), and σ_i^2 is per-edge aleatoric variance. An uncertainty-weighted causal attention assigns relevance,

$$\alpha_i = \text{softmax}_i\left(-\frac{\|\Delta p_i\|_2^2}{\sigma_i^2 + \varepsilon}\right), \quad z_t = \sum_{i=1}^M \alpha_i\, W_e\, \mathbf{e}_i^t, \tag{7}$$

with small $\varepsilon > 0$ for numerical stability and W_e a learnable projection. This prioritizes nearer and more confident actors while remaining fully differentiable.

The decision state concatenates the attended interaction embedding with control-relevant scalars,

$$s_t = \left[z_t;\; v_{\text{ego}};\; a_{\text{ego}}^{t-1};\; d_{\text{goal}};\; \phi_{\text{lane}};\; \sigma_{\text{ale}}^2\right]. \tag{8}$$

Here v_{ego} is ego speed, a_{ego}^{t-1} the previous action for smoothness, d_{goal} route progress (arc-length to the next target on $\mathcal{R}$), ϕ_{lane} lane-centerline features at p_t (curvature, heading offset consistent with κ_i), and σ_{ale}^2 aggregated aleatoric uncertainty at decision time. Lane-adherence variables (d_L, μ_A) are computed as in Sec. 3.2 and used by reward and penalties, but are not required in s_t at every update. As shown in Fig. 1 (a), z_t fuses ego dynamics, lane geometry, progress, and aleatoric variance to form s_t, exposing edge-wise influences and confidence to $\pi_\theta(a|s)$ and letting gradients capture safety-critical interactions.

3.4. Dense Multi-Objective Shaping

This subsection specifies the dense reward that operationalizes the MDP and constraints (Secs. 3.1–3.2), replacing sparse events with smooth surrogates for stable gradients across towns, weather, and dynamics. The shaped reward comprises safety, progress, comfort, and uncertainty terms, defined as *differentiable multi-objective reward shaping*.

The per-step reward is a convex combination,

$$r_t \;=\; w_s r_s \;+\; w_p r_p \;+\; w_c r_c \;+\; w_u r_u, \tag{9}$$

with nonnegative weights $w_\cdot$ satisfying $\sum_\cdot w_\cdot = 1$, used as r_{d} in the Lagrangian objective of Eq. (5). All components are continuous, bounded, and differentiable to match clipped actor-critic updates. Unlike prior formulations treating uncertainty as a standalone penalty [4, 27], the uncertainty term uses the same $\bar{\sigma}$ that drives relational attention in Eq. (7) and gates policy entropy (Sec. 3.5), making the reward part of the unified confidence interface.

Lane adherence, proximity, and signal obedience are encoded with smooth surrogates consistent with Eqs. (2)–(3). Let d_L be lateral deviation with soft barrier $\psi_L(d_L, \mu_A) = \tanh(d_L/\tau_d)$, where corridor membership μ_A shrinks admissible width, $i^\star$ the nearest actor at $\mathrm{dist}(p_t, p_t^{(i^\star)})$, with proximity surrogate $\psi_P = \exp(-\mathrm{dist}/\tau_p)$, and $\rho_t \in [0,1]$ red-light or stop-line noncompliance. The safety term is

$$r_s \;=\; 1 \;-\; \kappa_L\,\psi_L(d_L,\mu_A) \;-\; \kappa_P\,\psi_P \;-\; \kappa_R\,\rho_t, \quad (10)$$

with nonnegative coefficients $\kappa_{\{L,P,R\}}$.

Route advancement follows the arc-length coordinate from Sec. 3.2 for progress. Let $\Delta s_t = s_{t+1}^{\mathrm{arc}} - s_t^{\mathrm{arc}}$ denote step progress, and let $\hat{v}_t$ be the projection of ego velocity onto the route tangent. A bounded progress reward is

$$r_p \;=\; \tanh(\Delta s_t/\tau_s) \quad \text{or} \quad r_p \;=\; \tanh(\hat{v}_t/\tau_v), \quad (11)$$

with scales $\tau_s, \tau_v > 0$ ensuring numerical stability.

Longitudinal jerk j_t and steering rate $\dot{\delta}_t$ are penalized quadratically to enforce comfort bounds from Sec. 3.2,

$$r_c \;=\; -\,\kappa_j\,j_t^2 \;-\; \kappa_\delta\,\dot{\delta}_t^2, \quad (12)$$

where $j_t = \|a_t^{\mathrm{veh}} - a_{t-1}^{\mathrm{veh}}\|_2/\Delta t$ and $\dot{\delta}_t = (a_t^{\mathrm{str}} - a_{t-1}^{\mathrm{str}})/\Delta t$.

Decision-time uncertainty is summarized by $\bar{\sigma} \in [0,1]$ and estimated in Sec. 3.5. A simple, calibrated form encourages confident operation while remaining smooth:

$$r_u \;=\; 1 \;-\; \bar{\sigma}, \quad (13)$$

with optional temperature scaling by validation. This term complements safety shaping without duplicating penalties. Fig. 1 (b) illustrates the four smooth components and their bounded ranges that stabilize optimization and transfer.

3.5. Uncertainty-Gated Exploration

This subsection formalizes decision-time uncertainty for risk-aware actor-critic training. Prior designs estimate one source or treat it diagnostically, leaving *policy entropy decoupled* from confidence and degrading safety. The goal is to jointly estimate both components, propagate through the ego-relational state, and gate entropy in Eq. (5).

Decision-time variance is decomposed as

$$\sigma_{\mathrm{dec}}^2 \;=\; \sigma_{\mathrm{ale}}^2 \;+\; \sigma_{\mathrm{epi}}^2. \quad (14)$$

Aleatoric variance σ_{ale}^2 is estimated per edge in Eq. (6) and injected into the attention of Eq. (7), reducing the weight of distant or noisy actors. Epistemic variance σ_{epi}^2 is obtained from an ensemble of critics $\{Q_{\phi_k}\}_{k=1}^K$ via $\sigma_{\mathrm{epi}}^2(s_t,a_t) = \mathrm{Var}_k[Q_{\phi_k}(s_t,a_t)]$, using the same s_t as in Eq. (8). A normalized summary $\bar{\sigma} \in [0,1]$ is produced by temperature-scaled min-max or logistic calibration on a held-out split, yielding a stable confidence signal across domains.

Uncertainty modulates exploration via an entropy term,

$$\mathcal{L}_{\mathrm{ent}} \;=\; -\beta(\bar{\sigma})\,H(\pi_\theta(\cdot|s_t))\,, \quad \beta(\bar{\sigma}) = \beta_0(1-\bar{\sigma}), \quad (15)$$

so low confidence ($\bar{\sigma}\uparrow$) reduces entropy and risk-taking, while high confidence restores exploration. This coupling links the relational state, uncertainty, and policy stochasticity, complementing r_u in Eq. (9) without duplicating penalties. The critic ensemble operates during training only; at inference, a single actor pass adds no runtime overhead.

The training objective augments the standard actor-critic loss $\mathcal{L}_{\mathrm{RL}}$ with the entropy gate and Lagrangian shaping:

$$\min_{\theta,\phi} \;\; \mathcal{L}_{\mathrm{RL}}(\theta,\phi;\, r_{\mathrm{d}}) \;+\; \lambda_{\mathrm{ent}}\,\mathcal{L}_{\mathrm{ent}},$$

where r_{d} is given by Eqs. (9)–(13) and penalties enter via Eq. (5). Fig. 1 (c) summarizes the flow from per-edge σ_{ale}^2 and ensemble σ_{epi}^2 to $\bar{\sigma}$ and entropy gating.

3.6. Causal-Uncertainty Transfer Learning

This subsection addresses policy transfer by aligning control-level semantics and confidence, as adapting only perception features leaves *causal-semantic alignment* and *uncertainty statistics* mismatched at the policy layer. The aim is a compact objective matching policy distributions, uncertainty-weighted attention, and uncertainty moments with meta-initialization for few-shot adaptation.

Let π_{θ_s} and π_{θ_t} be source and target policies consuming state s_t from Eq. (8), $\boldsymbol{\alpha}_d$ the attention vector from Eq. (7) in domain $d \in \{s,t\}$, and u_d uncertainty moments of $\bar{\sigma}$ and per-edge σ_i^2. Define $\mathcal{L}_{\mathrm{KL}} = \mathrm{KL}\big(\pi_{\theta_s}(\cdot|s_t)\,\|\,\pi_{\theta_t}(\cdot|s_t)\big)$. The causal-uncertainty transfer loss is

$$\begin{aligned}\mathcal{L}_{\mathrm{trans}} = \mathcal{L}_{\mathrm{KL}} + \lambda_\alpha\,\mathrm{MMD}\big(\boldsymbol{\alpha}_s,\boldsymbol{\alpha}_t\big) \\ + \lambda_u\,\|u_s - u_t\|_2^2,\end{aligned} \quad (16)$$

aligning action distributions, attention, and uncertainty statistics for risk-aware behavior. The MAML step runs across source domains during training only; at inference, π_{θ_t} requires no ensemble or meta-gradient computation.

Initialization applies a MAML-style step to accelerate few-shot adaptation under Eqs. (5) and (15):

$$\theta^\star \;=\; \arg\min_\theta \sum_{d\in\mathcal{D}} \mathcal{L}_{\mathrm{RL}}^{(d)}\Big(\theta - \alpha\,\nabla_\theta \mathcal{L}_{\mathrm{RL}}^{(d)}(\theta)\Big)\,, \quad (17)$$

after which a few target updates refine π_{θ_t} while preserving aligned attention and uncertainty profiles.

The full training objective augments actor-critic learning with entropy gating and transfer regularization:

$$\min_{\theta,\phi} \;\; \mathcal{L}_{\mathrm{RL}}(\theta,\phi;\, r_{\mathrm{d}}) + \lambda_{\mathrm{ent}}\,\mathcal{L}_{\mathrm{ent}} + \lambda_T\,\mathcal{L}_{\mathrm{trans}}, \quad (18)$$

where r_{d} is defined in Sec. 3.4 and $\mathcal{L}_{\mathrm{ent}}$ in Eq. (15). This ensures control-relevant semantics and confidence calibration remain consistent across towns, traffic, and weather. Fig. 1 (d) summarizes causal-uncertainty alignment from source to target with meta-initialization for adaptation.

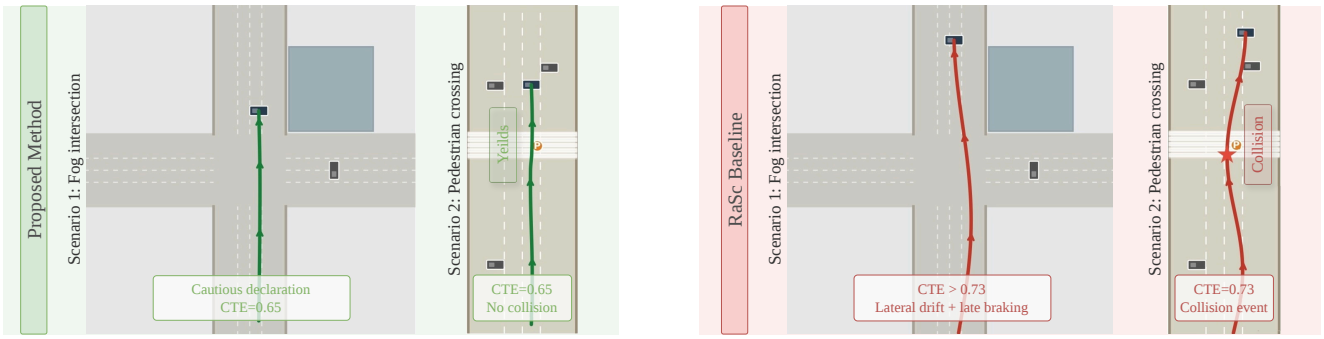


Figure 2. Trajectory comparison on Town10HD. Proposed method (green, left) vs. RaSc [8] (red, right). (a) Fog intersection: cautious deceleration (CTE = 0.65) vs. lateral drift (CTE > 0.73). (b) Pedestrian crossing: no collision (CTE = 0.65) vs. collision event (CTE = 0.73).

4. Closed-Loop Validation and Assessment

This section validates the framework across towns, traffic densities, and weather, linking outcomes to design choices in Secs. 3.3–3.6. Evaluation covers safety, stability, and generalization against ST-P3 [10], ThinkTwice [14], TransFuser [5], and RaSc [8], all retrained under identical CARLA 0.9.15 conditions. Results are reported for the ego-relational state, reward shaping, uncertainty-gated exploration, policy transfer, and a consolidated ablation.

4.1. Simulation Protocol and Metrics

Experiments are conducted in CARLA 0.9.15 with synchronous stepping at 20 Hz with a Tesla Model 3 ego vehicle using SAC with the entropy-gated objective in Sec. 3.5 and Lagrangian shaping in Eq. (5). Training is conducted in Town10HD under a deliberately adverse regime: heavy rain and dense fog at night (cloudiness 90%, precipitation 90%, fog density 40%, sun altitude $-25°$), with NPC traffic drawn from a moderate-to-high range per episode.

Hyperparameters follow standard SAC practice: replay buffers of 2×10^5 transitions, batch size 512, discount factor $\gamma = 0.99$, target update coefficient $\tau = 5 \times 10^{-3}$, Adam with learning rate 3×10^{-4}, and $\beta_0 \in \{0.5, 1.0\}$ selected on a validation split via $\beta(\bar{\sigma}) = \beta_0(1 - \bar{\sigma})$. Training runs for 5×10^5 steps with evaluation every 10k steps. Inference runs at 20 Hz on a single NVIDIA RTX 3090; the per-step actor inference with top-K aggregation adds ≤ 3 ms (batch size 1) over a vanilla SAC baseline, confirming real-time feasibility; the critic ensemble and MAML loop are training-only and contribute no overhead at deployment.

Closed-loop statistics follow the definitions in Sec. 3.2. Collision, off-road, and timeout rates are measured per kilometer as $\mathrm{q/km} = N_\mathrm{q}/D_\mathrm{km}$, where D_km is the total distance traveled in kilometers and N_q is the count of infractions of type $q \in \{\mathrm{coll}, \mathrm{off}, \mathrm{to}\}$. Route completion is computed as $\mathrm{RC}(\%) = 100 \cdot (\text{traversed arc-length}/\text{route arc-length})$. The success rate is defined as $\mathrm{SR} = N_\mathrm{succ}/N_\mathrm{ep}$.

Driving Score (DS) in Tab. 1 summarizes overall closed-loop driving performance. The Infraction Score, $\mathrm{IS} = \prod_{q \in \{\mathrm{coll},\mathrm{off},\mathrm{to}\}} \left(1 - \min\left(1, \frac{\mathrm{q/km}}{\kappa_q}\right)\right)$ with $\kappa_q = 0.02$ captures the safety penalty independently. Driving stability is evaluated using Cross-Track Error (CTE) and heading error in Eq. (2), reflecting the lane variables shaped in Sec. 3.4.

4.2. Validation of Ego-Relational State

The first set of experiments evaluates the causal ego-relational state in Sec. 3.3 against baselines differing in state encoding: ST-P3 [10] and ThinkTwice [14] rely on spatial-temporal tensors, TransFuser [5] fuses BEV and camera features, and RaSc [8] augments imitation with risk awareness but employs global maps. The proposed state replaces global tensors with an ego-node relational graph encoding relative kinematics, lane geometry, semantics, and per-entity aleatoric variance, aggregated via uncertainty-weighted attention in Eq. (7) into s_t in Eq. (8).

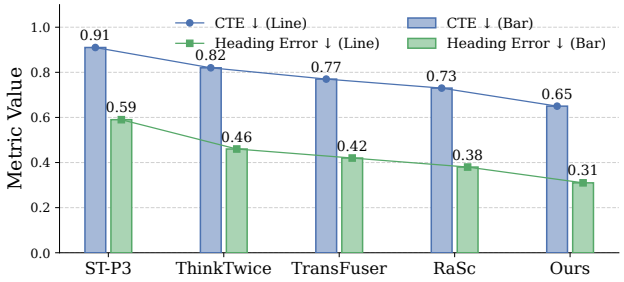


Figure 3. Ego-relational state on Town10HD: CTE and heading error reduced by uncertainty-weighted attention versus baselines.

Fig. 3 reports CTE and heading error on Town10HD. The causal state reduces lateral deviation compared to tensor and BEV-based encoders. Average CTE drops to 0.65 (28.6% below ST-P3, 20.7% below ThinkTwice, 15.6%/11.0% over TransFuser/RaSc), and heading error to 0.31 (47.5% from ST-P3, 32.6%, 26.2%, 18.4% over ThinkTwice, TransFuser, RaSc), reflecting the fusion of lane geometry and ego dynamics with uncertainty-weighted features in Eq. (8).

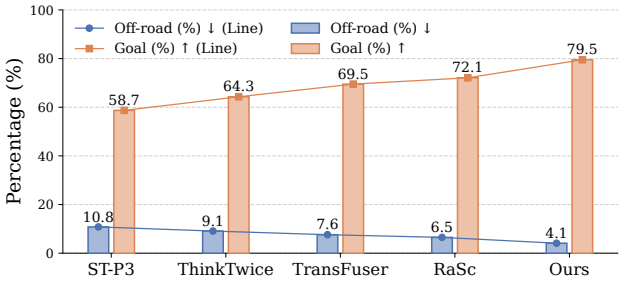


Figure 4. Route completion metrics. Off-road (%) decreases and goal rate increases when the policy uses the causal state.

Fig. 4 shows route completion via off-road percentage and goal rate. Off-road drops from 10.8% (ST-P3) to 4.1% (62% reduction), with 55%, 46%, and 37% gains over ThinkTwice, TransFuser, and RaSc. Goal completion rises to 79.5%, improving 35.4% over ST-P3, 23.6% over ThinkTwice, and 14.4%/10.3% over TransFuser and RaSc.

These trends show that the uncertainty-weighted relational state improves stability and goal attainment, reducing off-road and boosting success compared to tensor baselines.

4.3. Impact of Dense Reward Shaping

The second set of experiments studies the dense multi-objective shaping in Sec. 3.4. The baseline methods use progress-dominated or event-driven rewards, with discontinuous penalties for collisions and lane departures. The proposed formulation replaces such thresholds with differentiable surrogates for safety, progress, comfort, and uncertainty, as summarized in Eqs. (9)–(13), so that gradients remain informative even under rare safety events.

Fig. 5 compares average episode rewards on Town10HD. Baselines achieve 156.4, 164.1, 174.2, and 182.9 for ST-P3, ThinkTwice, TransFuser, and RaSc respectively. The proposed framework reaches 265.3, improving by 69.6%, 61.7%, 52.3%, and 45.1% over baselines. Learning curves show smoother growth and fewer oscillations, indicating differentiable penalties in Eq. (10) and comfort regularization in Eq. (12) reduce sensitivity to large-magnitude events.

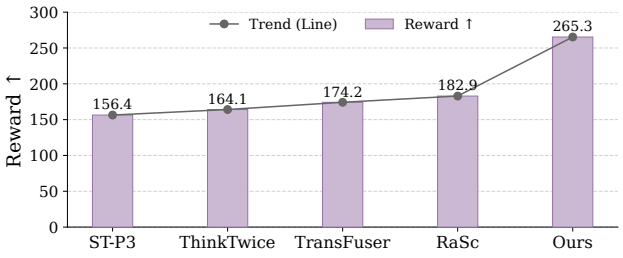


Figure 5. Reward comparison where differentiable multi-objective shaping yields higher, more stable returns than baselines.

Reward gains improve safety and comfort in a closed loop. Continuous surrogates in r_s and r_p let the critic adapt even in near-miss cases. Jerk and steering-rate penalties (Eq. 12) reduce oscillations, lowering CTE and heading error. The uncertainty term r_u (Eq. 13) promotes well-observed states, making rewards robust to sensor noise.

4.4. Uncertainty-Gated Exploration

The third set of experiments isolates the impact of joint uncertainty estimation and entropy gating in Sec. 3.5, where baselines ignore uncertainty or treat it diagnostically without feedback to attention or exploration. The proposed design propagates aleatoric variance through attention, estimates epistemic variance via a critic ensemble, and compresses both into $\bar{\sigma}$ for the entropy gate in Eq. (15).

Fig. 6 reports exploration variance, collision rate, and stability. The uncertainty-gated policy reduces variance to 0.62 vs. 0.89, 0.80, 0.72, 1.00 for ThinkTwice, TransFuser, RaSc, and ST-P3 (38%, 30.3%, 22.5%, 13.9% reductions), driven by the entropy penalty in Eq. (15) suppressing randomness under high $\bar{\sigma}$, especially under fog or occlusion.

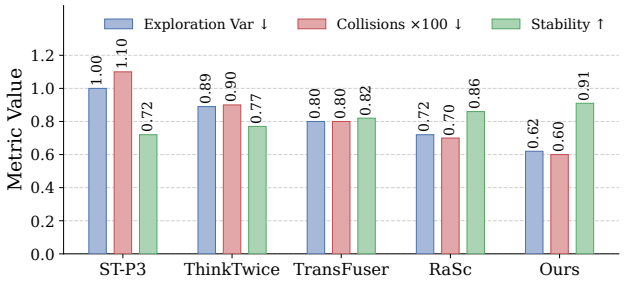


Figure 6. Uncertainty-aware exploration and safety metrics.

Collision rates drop from 0.011 (ST-P3), 0.009 (ThinkTwice), 0.008 (TransFuser), and 0.007 (RaSc) to 0.006 with uncertainty-gated exploration, 45.5%-14.3% fewer than baselines. Stability rises to 0.91, exceeding ST-P3 (0.72), ThinkTwice (0.77), TransFuser (0.82), and RaSc (0.86).

Results validate uncertainty-exploration coupling: variance and critic ensemble regulate attention and entropy, making the agent cautious under ambiguity and exploratory when confident. Collision and stability gains show improved safety-exploration balance over fixed schedules.

4.5. Impact of Causal-Uncertainty Transfer

The final experiments study policy transfer from Town10HD to Town02 and Town05 using the causal-uncertainty objective in Sec. 3.6. Three scenarios are reported: (i) cross-town transfer to Town02; (ii) zero-shot transfer to Town05 under mixed weather; and (iii) source-domain performance on Town10HD as reference.

Tab. 1 reports Town02 transfer averaged over 20 episodes. Driving Score rises from 188.6 (policy learning)

Table 1. Closed-loop results across towns. Town02 values averaged over 20 episodes; Town05 and Town10HD over 20 episodes each.

Domain	**Variant**	SR (%)	RC (%)	DS	IS	Coll./km	OR/km	TO/km
Town10HD (source)	Source agent	91.2	94.1	94.1	1.00	0.000	0.000	0.000
Town05 (zero-shot)	Source agent	**100.0**	94.6	94.6	1.00	0.000	0.000	0.000
Town02 (cross-town)	Policy learning	72.1	75.2	188.6	0.88	0.007	0.005	0.003
	Source domain	80.3	82.6	205.7	0.92	0.006	0.004	0.002
	Target (ours)	85.0	84.1	**214.3**	**0.94**	**0.005**	**0.003**	**0.001**

to 205.7 (source domain) and 214.3 (full transfer). Route completion improves from 75.2% to 84.1%, collisions/km fall from 0.007 to 0.005, and Infraction Score increases from 0.88 to 0.94, confirming that aligning action distributions, attention, and uncertainty at the decision layer outperforms target-only training and perception adapters [13, 20].

The Town05 zero-shot result (SR 100%, CTE 0.192 m, heading error 0.021 rad) confirms the causal-uncertainty interface generalizes without target-domain adaptation to a topologically distinct map and mixed weather. Town10HD yields 1,693 episodes (reward 265.3, CTE 0.65, heading error 0.31); Town02 (20 episodes) achieves reward 558.9, CTE 0.73, heading error 0.18, with causal-semantic alignment in Eq. (16) and meta-initialization in Eq. (17) stabilizing attention and uncertainty across domains.

Fig. 2 compares closed-loop ego trajectories on Town10HD under two safety-critical scenarios: (1) a fog-occluded intersection and (2) a pedestrian-crossing in dense traffic. RaSc [8] exhibits lateral drift and delayed braking in both cases, resulting in higher CTE and a collision in scenario (2). As $\bar{\sigma}$ rises under occlusion, the entropy-gating in Eq. (15) suppresses policy stochasticity, causing the proposed method to decelerate earlier and maintain tighter lane adherence, avoiding collision in both scenarios.

Table 2. Component-wise ablation on Town10HD and Town02.

Variant	CTE↓	Coll./km↓	DS (T02)↑	Stab.↑
w/o Unc. Attn.	0.76	0.008	203.5	0.84
w/o Ensemble	0.68	0.009	208.1	0.87
w/o Ent. Gate	0.71	0.008	206.7	0.86
Event Reward	0.74	0.009	195.9	0.83
w/o Transfer	0.65	0.006	194.1	0.91
w/o MAML	0.65	0.006	200.8	0.91
Full model (ours)	**0.65**	**0.006**	**214.3**	**0.91**

4.6. Ablation Study

Tab. 2 quantifies each module contribution on Town10HD. Removing uncertainty-weighted attention (*w/o Unc. Attn.*) raises CTE by 0.11 and heading error by 0.07, confirming confidence-aware prioritization is essential for stability. Disabling critic ensemble (*w/o Ensemble*) increases collisions by 0.003/km; removing entropy gating (*w/o Ent. Gate*) raises variance by 21% and collisions by 0.002/km, validating $\bar{\sigma}$-epistemic-stochasticity coupling.

Replacing the differentiable reward with event-driven penalties (*Event Reward*) drops DS by 18.4 points and raises CTE by 0.09, consistent with optimization instability in Sec. 3.4. Removing the transfer objective (*w/o Transfer*) and MAML (*w/o MAML*) reduce Town02 DS to 194.1 and 200.8, isolating their complementary contributions.

Hyperparameter sensitivity is assessed using $\beta_0 = 1.0$, $w_s = 0.4$, $w_p = 0.3$, $w_c = w_u = 0.15$, $\lambda_\alpha = 0.1$, $\lambda_u = 0.05$: varying $\beta_0 \in \{0.5, 1.0\}$ changes collision rate by $\leq$0.001/km and CTE by 0.03; shifting weights toward progress ($w_p = 0.5$) raises completion by 2.1% but increases collisions by 0.002/km; sweeping $\lambda_\alpha, \lambda_u \in \{0.05, 0.1, 0.2\}$ yields Town02 DS variation below 3.5 points.

Overall, the ego-relational state improves stability, multi-objective reward reduces collisions, uncertainty-gated exploration ties to confidence, and causal-uncertainty transfer boosts scores across towns. Future work will extend to Sim2Real transfer, interventional causal modeling, and reduced ensemble overhead, with evaluation on the CARLA Leaderboard and Bench2Drive [15] as planned.

5. Conclusion

A unified Deep RL framework built around a control-layer uncertainty interface was presented, coupling a causal ego-relational state, differentiable multi-objective reward, joint aleatoric-epistemic estimation, entropy-gated exploration, and causal-uncertainty transfer. Closed-loop evaluation in CARLA across adverse weather, dense traffic, and cross-town settings demonstrated consistent gains in stability, driving score, and infraction rates over strong E2E baselines, confirming that uncertainty-weighted interaction structure benefits control-level reasoning beyond what perception-centric encoders provide. Uncertainty-gated entropy reduced collisions without sacrificing route completion, zero-shot transfer confirmed generalization without target-domain adaptation, and causal-uncertainty alignment improved few-shot transfer by matching action distributions, attention patterns, and uncertainty statistics at the decision layer. Evaluation on standardized benchmarks and real-vehicle deployment remain directions for future work.

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